

4.1 Vector Space

Vector space and axioms:

Vector Space:

A vector space V over a field T of elements $\{\alpha, \beta, \gamma, \dots\}$, called scalars, is a set of elements $\{\underline{u}, \underline{v}, \underline{w}, \dots\}$, called vectors, such that following axioms are satisfied:

1. Vector operation: $\underline{u} + \underline{v}$, such that for all $\underline{u}, \underline{v}, \underline{w} \in V$

AI, Closure: $\underline{u} + \underline{v} \in V$

AII, Associativity: $(\underline{u} + \underline{v}) + \underline{w} = \underline{u} + (\underline{v} + \underline{w})$

AIII, Zero: There exists a zero (null) vector $\underline{0} \in V$ such that $\underline{v} + \underline{0} = \underline{v}$

AIV, Negative: There exists an negative $-\underline{u} \in V$ such that $\underline{u} + (-\underline{u}) = \underline{0}$

2. Scalar multiplication: $\alpha \underline{u}$, such that for all $\underline{u}, \underline{v} \in V$, and all $\alpha, \beta \in T$ (typically $\mathbb{R}$)

MI, Closure: $\alpha \underline{u} \in V$

MII, Associativity: $(\alpha \beta) \underline{u} = \alpha (\beta \underline{u})$

MIII, Distributivity: a) $(\alpha + \beta) \underline{u} = \alpha \underline{u} + \beta \underline{u}$
b) $\alpha (\underline{u} + \underline{v}) = \alpha \underline{u} + \alpha \underline{v}$

MIV, Unitary: $1 \in T$, $1 \underline{u} = \underline{u}$
typically $\mathbb{R}$

4.2 Examples of Vector Spaces.

- $A \in \mathbb{R}^{m \times n}$ ($m \times n$, m rows, n columns) Matrix

- $A \in \mathbb{R}^n$ (or $\mathbb{R}^n$) column (or row) matrix

- Polynomials

4.3 Theorems and results

Cancellation theorem: $\underline{x} + \underline{u} = \underline{y} + \underline{u} \implies \underline{x} = \underline{y}$

Definition of subtraction:

$$\underline{u} - \underline{v} = \underline{u} + (-\underline{v})$$

Propositions: a) $-\underline{u} + \underline{u} = \underline{0}$

b) $\underline{0} + \underline{u} = \underline{u}$

c) $-\underline{u}$ of $\underline{u}$ is unique

d) $-(-\underline{u}) = \underline{u}$

Commutativity:

$$\underline{u} + \underline{v} = \underline{v} + \underline{u}$$

4.4 Properties of Vector Spaces

Properties of zero $\underline{0}$: a.) $\underline{0}x = \underline{0}$

b.) $\alpha \underline{0} = \underline{0}$

c.) iff $\alpha x = \underline{0}$, either $\alpha = 0$ or $x = \underline{0}$

d.) $\underline{0}$ is unique

More proposition: a.) $(-\alpha)x = -(\alpha x) = \alpha(-x)$

↳ i.) $(-1)x = -x$

5.1 Subspaces

Definition of a subspace: A subset of vector space V is a subspace iff U is a vector space over the same field with same vector addition and scalar multiplication.

Subspace test: Let U be a subset of vector space V . Then U is a subspace of V , over the same field T with the same vector addition and scalar multiplication as V , iff for all $u, v \in U$ and $\alpha \in T$,

SI, Zero: There exists a $\underline{0} \in U$ $\leftarrow$ Must be same $\underline{0}$ as V

SII, Closure under vector addition: $u + v \in U$

SIII, Closure under scalar multiplication: $\alpha u \in U$

5.2 Examples of subspaces

Image space: Given A is an arbitrary matrix $\in {}^m\mathbb{R}^n$
 $\text{im } A \triangleq \{y \mid y = Ax, x \in {}^n\mathbb{R}\} \subseteq {}^m\mathbb{R}$

y , m row column vector

In understandable words: Image space of A , a $m \times n$ matrix is the column to the system $y = Ax$ for every x , n row column vector

$$(m \times 1) = (m \times n)(n \times 1)$$

$$\begin{bmatrix} j \\ k \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

SI: take $x = \underline{0}$
 $Ax = \underline{0}$
 $y = \underline{0}$

SII: $y_1 + y_2$
 $= Ax_1 + Ax_2$
 $= A(x_1 + x_2)$

$\therefore x_1 + x_2 \in {}^n\mathbb{R}$
 $\therefore A(x_1 + x_2) \in \text{im } A$

SIII: $\alpha y = \alpha(Ax)$
 $= A(\alpha x)$

$\therefore \alpha x \in {}^n\mathbb{R}$

$\therefore A(\alpha x) \in \text{im } A$

Null (or solution) space: $\text{null } A \triangleq \{x \mid Ax = 0\} \subseteq \mathbb{R}^n$ for $A \in \mathbb{R}^{m \times n}$

i.e. x is the solution of the system $Ax = 0$

SI: $x = 0$ is always
a solution, $0 \in \text{null } A$

SII: if $x_1 + x_2$ is a solution for
 $A(x_1 + x_2) = Ax_1 + Ax_2 = 0 + 0 = 0$
so $x_1 + x_2 \in \text{null } A$

SIII: if x is a solution,
 $\alpha Ax = \alpha(0) = 0$
so $\alpha x \in \text{null } A$

Subspace of functions: Consider the set of functions, a subset of $\mathcal{F}[0, 1]$, in which $f(0) = 0$
that is, $\mathcal{U} = \{f \mid f(0) = 0\} \subset \mathcal{F}[0, 1]$

SI: $z \in \mathcal{F}[0, 1]$ is contained
in $\mathcal{U}$ for $z(0) = 0$

SII: $f_1 + f_2 \in \mathcal{U}$, then $f_1 + f_2 \in \mathcal{U}$
 $(f_1 + f_2)(0) = f_1(0) + f_2(0)$
 $= 0 + 0 = 0$

SIII: $f \in \mathcal{U}$ then $\alpha f \in \mathcal{U}$
 $(\alpha f)(0) = \alpha(f(0)) = \alpha \cdot 0$
 $= 0$

Polynomial Space: $\mathcal{P} = \{p \mid p \in P_n, p(0) = 0\} \subset P_n$

Similar proof

5.3 Spanning Sets

Definition of a linear combination: A vector $v \in V$ is a linear combination of $\{v_1, v_2, \dots, v_n\} \subset V$ iff it can be expressed as

$$v = \sum_{j=1}^n \lambda_j v_j + \lambda_2 v_2 + \dots + \lambda_n v_n$$

for some $\lambda_j \in \mathbb{T}$ (typically $\mathbb{R}$)

Definition of a span: The span of $\{v_1, v_2, \dots, v_n\} \subset V$, denoted as $\text{span}\{v_1, v_2, \dots, v_n\}$ is

$$\text{span}\{v_1, v_2, \dots, v_n\} = \left\{ v \mid v = \sum_{j=1}^n \lambda_j v_j, \lambda_j \in \mathbb{T} \right\}$$

Propositions from span and linear combinations:

a) $\text{span}\{v_1, v_2, \dots, v_n\} \subset V$ is a subspace of vector space V .

b) Let $\mathcal{U} = \text{span}\{v_1, v_2, \dots, v_n\}$, a subspace of (or equal to) vector space V .
If W is a subspace of V containing vectors $\{v_1, v_2, \dots, v_n\}$, then $\mathcal{U}$ is a subspace of W .

Between 5.3 and 6.1

Intersection: The intersection of 2 sets U and W is the set
 $U \cap W = \{x \mid x \in U \text{ and } x \in W\}$

Theorem: If U and W are subspaces of vector space V , then $U \cap W$ is also a subspace of V

Sum: The sum of 2 sets U and W is the set
 $U + W = \{u + w \mid u \in U \text{ and } w \in W\}$

Theorem: If U and W are subspaces of vector space V , then $U + W$ is also a subspace of V

Alternative subspace test:
a.) show V is non-empty
b.) show $cx + y \in V$

Equivalence of Sets: $U = V$ if
1.) $U \subseteq V$ and
2.) $V \subseteq U$

Equivalence of Spans: $\text{span}\{v_1, v_2, \dots, v_m\} = \text{span}\{u_1, u_2, \dots, u_n\}$

if 1.) $\text{span}\{v_1, v_2, \dots, v_m\} \subseteq \text{span}\{u_1, u_2, \dots, u_n\}$

i.e. v_i can be expressed as $v_i = a_1 u_1 + a_2 u_2 + \dots + a_n u_n$

2.) $\text{span}\{u_1, u_2, \dots, u_n\} \subseteq \text{span}\{v_1, v_2, \dots, v_m\}$

Example: $\text{span}\{u, v\} = \text{span}\{u + v, u - v\}$

6.1 Linear independence

Linear independence: A set of vectors $\{v_1, v_2, \dots, v_n\} \subset V$ is linearly independent iff:

$$\sum_{j=1}^n \lambda_j v_j = \lambda_1 v_1 + \dots + \lambda_n v_n = 0$$

implies that all $\lambda_j = 0$.

Proposition I: If $\{v_1, v_2, \dots, v_n\} \subset V$ is linearly independent and $v = \sum_{j=1}^n \lambda_j v_j$
Then for all $v \in V$, λ_j are uniquely determined.

Minimal Spanning Sets

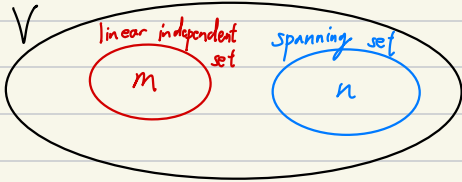
Let $\{v_1, v_2, \dots, v_n\} \subset V$, a vector space. For every v_k ($k = 1 \dots n$), $\text{span}\{v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_n\} \subsetneq \text{span}\{v_1, \dots, v_n\}$ iff $\{v_1, v_2, \dots, v_n\}$ is linearly independent.

strict
subset

Corollary: Let $\{v_1, v_2, \dots, v_n\} \subset V$, a vector space. For at least one v_k ($k=1, \dots, n$),
 $\text{span}\{v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_n\} = \text{span}\{v_1, \dots, v_n\}$ iff $\{v_1, v_2, \dots, v_n\}$ is linearly dependent.

Fundamental Theorem of Linear Algebra:

Let V be a vector space spanned by n vectors, if a set of m vectors from V is linearly independent, then $m \leq n$



Replacement Lemma: Let V be a non-zero vector space and let $\{x_1, x_2, \dots, x_k\}$ span V , let α be a non-zero vector in V , where

$$\alpha = c_1 x_1 + c_2 x_2 + \dots + c_k x_k$$

If $c_j \neq 0$ for some $j=1, 2, \dots, k$ then the set with x_j replaced by α spans V , i.e.

$$\text{span}\{x_1, x_2, \dots, x_{j-1}, x_{j+1}, \dots, x_k, \alpha\} = V$$

6.2 Bases

Base: A set of vectors $\{e_1, e_2, \dots, e_n\} \in V$ is a **basis** for the vector space V iff

1. $\{e_1, e_2, \dots, e_n\}$ is linearly independent, and
2. $\{e_1, e_2, \dots, e_n\}$ spans V

$\hookrightarrow$ Every non-zero vector space has infinitely many bases

Theorem: Every basis for a given vector space contains the same number of vectors

Dimension: The dimension of a vector space V , denoted $\dim V$, is the number of vectors in any of its bases.

Propositions from dim: Let V be a finite-dimensional vector space, $\dim V = n$

1. A linearly indep. set of vectors in V can contain at most n vectors
2. A spanning set for V must contain at least n vectors

6.4 Existence of Bases

Theorem: Let $\{v_1, v_2, \dots, v_n\} \subset V$ be linearly indep.

Then for a vector $v \in V$, $\{v, v_1, v_2, \dots, v_n\}$ is lin indep. iff $v \notin \text{span}\{v_1, v_2, \dots, v_n\}$.

Contraposition: $\{v, v_1, v_2, \dots, v_n\}$ is linearly dep. iff $v \in \text{span}\{v_1, v_2, \dots, v_n\}$

Existence of Bases: Let V be a vector space spanned by finite set of vectors.

Then every linearly indep. set of vectors in V can be extended to a basis for V .

1. Start with a set of vectors

$B_k = \{v_1 \dots v_k\} \subset V$, where $1 \leq k \leq m \rightarrow$ (FTLA)

2. Either B_k spans $V \rightarrow$ found basis

$\hookrightarrow B_k$ does not span $V \rightarrow$ There must exist some vector $v_{k+1} \in V$ and $v_{k+1} \notin \text{span } B_k$
 $B_{k+1} = \{v_1 \dots v_k, v_{k+1}\}$ is linearly independent.

3. Repeat Step 2 w/ $k+2, k+3, k+4 \dots$

4. Until we come to $k=m$, where B_m is a basis $\rightarrow$ If we continue, we contradict FTLA.

Theorem: Let U and W be subspaces of a finite-dimensional vector space V . Then:

- 1.) U is finite dimensional, $\dim U \leq \dim V$
- 2.) If $U \subseteq W$, then $\dim U \leq \dim W$
- 3.) If $U \subseteq W$ and $\dim U = \dim W$, then $U = W$

Theorem: Any spanning set for a vector space V contains a basis for V
 $\hookrightarrow$ Remove vectors to get a basis.

Theorem: Let V be a vector space and $\dim V = n$. Then

1. Any set $\{v_1 \dots v_n\} \subset V$ that is linearly independent is a basis for V ; and
2. Any set $\{v_1 \dots v_n\} \subset V$ that spans V is a basis for V

7.2 Row and Column Spaces

Representing A in col or row: If $A \in \mathbb{R}^{m \times n}$, $A = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix} = [c_1 \ c_2 \ \dots \ c_n]$

Row space: Row space of $A \in \mathbb{R}^{m \times n}$, denoted $\text{row } A$, is
 $\text{row } A \triangleq \text{span} \{r_1, r_2 \dots r_m\}$ where $r_i \in \mathbb{R}^n$ are the rows of A .

Column space: Column space of $A \in \mathbb{R}^{m \times n}$, denoted $\text{col } A$, is
 $\text{col } A \triangleq \text{span} \{c_1, c_2 \dots c_n\}$ where $c_j \in \mathbb{R}^m$ are the columns of A

$\hookrightarrow \text{col } A = \text{im } A = \{y \mid y = Ax, x \in \mathbb{R}^n\}$

Proposition for col./row space: Let $A \in \mathbb{R}^{m \times n}$, $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$. Then $\text{row } UA \subseteq \text{row } A$, equality holding ($\text{row } UA = \text{row } A$) if U is invertible.
And $\text{col } AV \subseteq \text{col } A$, with equality holding ($\text{col } AV = \text{col } A$) if V is invertible.
 $\text{row } UA = \text{row } A$ does not imply U is invertible, and $\text{col } AV = \text{col } A \dots$

Proposition for invertible matrix: Let $\{x_1, x_2 \dots x_n\} \subset \mathbb{R}^n$ and let $U \in \mathbb{R}^{n \times n}$ be invertible.
Then $\{x_1, x_2 \dots x_n\}$ is linearly indep. iff $\{Ux_1, Ux_2 \dots Ux_n\}$ is linearly indep.

(row space)

reduced row echelon form

Lemma I: Let $A \in \mathbb{R}^{m \times n}$. Then 1) $\text{row } \tilde{A} = \text{row } A$,
2) $\dim \text{row } \tilde{A} = \dim \text{row } A$,
3) nonzero rows of $\tilde{A}$ constitute a basis for $\text{row } A$.
 $\hookrightarrow$ Note $E_1 \cdots E_3 E_2 E_1 A = \tilde{A}$, E_k are invertible

(column space)

Lemma II: Let $A \in \mathbb{R}^{m \times n}$. Then 1) The set of columns with leading 1s, $\{\tilde{c}_{j_1}, \tilde{c}_{j_2}, \dots, \tilde{c}_{j_r}\}$ of $\tilde{A}$, the row-reduced form of A , constitutes a basis for $\text{col } \tilde{A}$ and
2) The corresponding columns $\{c_{j_1}, c_{j_2}, \dots, c_{j_r}\}$ of A constitute a basis for $\text{col } A$.
3) Hence $\dim \text{col } \tilde{A} = \dim \text{col } A$.

$\hookrightarrow$ In general, $\text{col } \tilde{A} \neq \text{col } A$

Row, column Theorem: Let $A \in \mathbb{R}^{m \times n}$, $\dim \text{row } A = \dim \text{col } A$

7.3 Rank of a Matrix

Rank: Let $A \in \mathbb{R}^{m \times n}$. The Rank of A , denoted $\text{rank } A$, is the common dim. of $\text{row } A$ and $\text{col } A$.

Properties of rank: If $A \in \mathbb{R}^{m \times n}$,

1) $\text{rank } A = \text{rank } \tilde{A}$

2) $\text{rank } A = \text{rank } A^T$

3) If $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ then

$\hookrightarrow \text{rank } UA \leq \text{rank } A$

$\hookrightarrow \text{rank } AV \leq \text{rank } A$

equality holding if U and V are invertible.

7.4 Dimension Formula

Dimension Formula: Let $A \in \mathbb{R}^{m \times n}$, $\dim \text{null } A = n - \text{rank } A$

$\hookrightarrow$ Also note for $Ax=0$ (or $\tilde{A}x=0$), # of free var. = # of unknowns - # of non-zero rows in $\tilde{A}$

Number of solutions to Linear system: - No Solution:

$\hookrightarrow \text{col } A \neq \text{col } [A|b]$

$\hookrightarrow \text{rank } A < \text{rank } [A|b]$

- One unique Solution:

$\hookrightarrow \text{col } A = \text{col } [A|b]$

$\hookrightarrow \text{rank } A = \text{rank } [A|b] = n$

$\hookrightarrow \dim \text{null } A = 0$, only $\vec{0}$

- Infinitely many solutions

$\hookrightarrow \text{col } [A|b] = \text{col } A$

$\hookrightarrow \text{rank } A = \text{rank } [A|b] < n$

$\hookrightarrow \dim \text{null } A > 0$

Equivalent Statements for square matrices:

Let $A \in \mathbb{R}^n$. The following statements are equivalent:

- 1.) A is invertible
- 2.) A has full rank n
- 3.) The rows of A are linearly independent
- 4.) The columns of A are linearly independent
- 5.) For $x \in \mathbb{R}^n$, $Ax = 0 \rightarrow x = \vec{0}$
- 6.) For $z \in \mathbb{R}^n$, $z^T A = 0 \rightarrow z = \vec{0}$

Random $\vec{0}$ Lemma:

Let $s \in \mathbb{R}^n$, if $s^T s = 0$, $s = \vec{0}$

Fredholm Alternative:

If $(A^T A)x = \vec{0} \rightarrow x = \vec{0}$,
then $A^T A$ is invertible

Equivalent Statements for full column rank matrices:

Let $A \in \mathbb{R}^m$. The following statements are equivalent:

- 1.) $\text{rank } A = n$
- 2.) The columns of A are linearly indep.
- 3.) For $x \in \mathbb{R}^n$, $Ax = 0 \rightarrow x = \vec{0}$
- 4.) $A^T A$ is invertible
- 5.) A has a "left inverse", i.e. $BA = I$ for some $B \in \mathbb{R}^n$

Equivalent Statements for full row rank matrices:

Let $A \in \mathbb{R}^m$. The following statements are equivalent:

- 1.) $\text{rank } A = m$
- 2.) The rows of A are linearly indep.
- 3.) For $z \in \mathbb{R}^m$, $z^T A = \vec{0} \rightarrow z = \vec{0}$
- 4.) AA^T is invertible
- 5.) A has a right inverse, i.e. $AB = I$ for some $B \in \mathbb{R}^n$

8.1 Linear Transformations

Linear transformation: The mapping $\mathcal{R}: V \rightarrow W$ is a linear transformation (operator) iff:

1.) Distributivity: $\mathcal{R}(u+v) = \mathcal{R}(u) + \mathcal{R}(v)$, for all $u, v \in V$

2.) Homogeneity: $\mathcal{R}(\lambda v) = \lambda \mathcal{R}(v)$ for all $\lambda \in \mathbb{F}$, $v \in V$

V = domain of $\mathcal{R}$

W = codomain / target space of $\mathcal{R}$

Trace Operator: $\text{tr } A = \sum_{i=1}^n a_{ii}$

$\hookrightarrow$ sum of all diagonal entries, returns a scalar

$\mathbb{R}^n \rightarrow \mathbb{R}$

Differential Operator: $D(p) = p'$
 $P_n \rightarrow P_{n-1}$

Identity Operator: $i(v) = v$
 $V \rightarrow V$

Zero Operator (Null Operator): $z(v) = \vec{0}$, $\vec{0}$ in W
 $V \rightarrow W$

Propositions for Linear Transformation: Let $\mathcal{R}: V \rightarrow W$ be a linear transformation.

1.) $\mathcal{R}(\vec{0}) = \vec{0}$

2.) $\mathcal{R}(-v) = -\mathcal{R}(v)$ for all $v \in V$

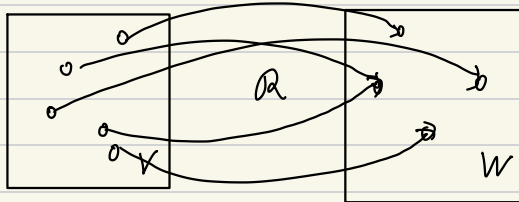
3.) For all $\lambda_i \in \mathbb{F}$, $v_i \in V$, $\mathcal{R}\left(\sum_{i=1}^k \lambda_i v_i\right) = \sum_{i=1}^k \lambda_i \mathcal{R}(v_i)$

8.2 Image and Kernels

Image: Image of linear transformation (operator) $\mathcal{R}: V \rightarrow W$, denoted $\text{im } \mathcal{R}$ is,
 $\text{im } \mathcal{R} = \{w \mid w = \mathcal{R}(v), v \in V\}$

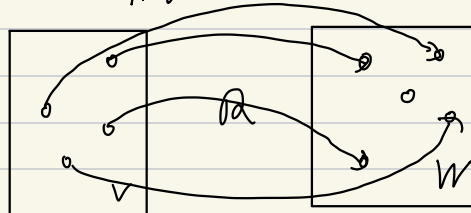
Kernel: Kernel of linear transformation (operator) $\mathcal{R}: V \rightarrow W$, denoted $\ker \mathcal{R}$ is,
 $\ker \mathcal{R} = \{v \in V \mid \mathcal{R}(v) = \vec{0}\}$
↖ in W

Surjective Mapping:



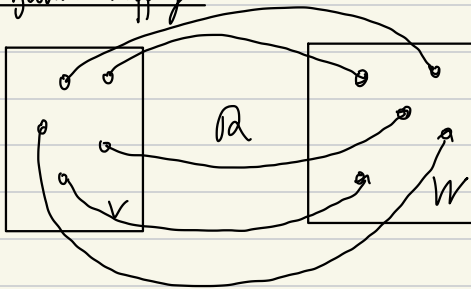
↳ Every element in W can be "reached by V " using $\mathcal{R}$
in $\mathcal{R} = W$

Injective Mapping:



↳ "One to One"
 $\mathcal{R}(v_1) = w$ and $\mathcal{R}(v_2) = w$
implies $v_1 = v_2$

Bijective mapping:



↳ "Covers all of W " and "One to One"
in $\mathcal{R} = W$
 $\mathcal{R}(v_1) = w$ and $\mathcal{R}(v_2) = w$
implies $v_1 = v_2$

Proposition of image kernel: Let $\mathcal{R}: V \rightarrow W$ be a linear transformation (operator).
Then $\text{im } \mathcal{R}$ is a subspace of W , $\ker \mathcal{R}$ is a subspace of V .

Dimension Formula: Let $\mathcal{R}: V \rightarrow W$ be any linear transformation (operator) with finite-dimensional $\text{im } \mathcal{R}$ and $\ker \mathcal{R}$.
Then, $\dim \ker \mathcal{R} + \dim \text{im } \mathcal{R} = \dim V$

8.3 Establishing One's Coordinates

Coordinates: Let $B = \{e_1, \dots, e_n\}$ be a basis for a vector space V over the field $\mathbb{R}$ (typically $\mathbb{R}$).
Then the coordinates of a vector $v \in V$ with respect to the basis B are scalars $v_1, \dots, v_n \in \mathbb{R}$ s.t.
 $v = v_1 e_1 + v_2 e_2 + \dots + v_n e_n$
is the unique linear combination of v in terms of B .
ex. In $\mathbb{R}_2$, if $B = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ then coordinates of $v = \begin{bmatrix} x \\ y \end{bmatrix}$ is (who wouldn't think), $\begin{bmatrix} x \\ y \end{bmatrix}$

Linear Transformation and Matrices

Let $\mathcal{R}: V \rightarrow W$, $\dim V = n$, $\dim W = m$,
some basis of V be $B_V = \{e_1, e_2, \dots, e_n\}$, some basis of W be $B_W = \{h_1, h_2, \dots, h_m\}$

Then, vectors in V can be "cast" as vectors in $\mathbb{R}^n$
vectors in W can be "cast" as vectors in $\mathbb{R}^m$ } *written in coordinates* $\rightarrow \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ and $\begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix}$

$\bar{w} \in W = \mathcal{R}(v) = \sum_{j=1}^n v_j \mathcal{R}(e_j) \rightarrow$ transforming basis then multiply by original coordinate.

Now, since $\mathcal{R}(v), \mathcal{R}(e_1), \dots, \mathcal{R}(e_n) \in W$, so we can express $\mathcal{R}(v)$ in terms of $\{h_1, \dots, h_m\}$.

$$\mathcal{R}(v) = \sum_{j=1}^m w_j h_j$$

$$\mathcal{R}(e_j) = \sum_{i=1}^m l_{ij} h_i, j=1, \dots, n \rightarrow \text{Each } e_n \text{ is a linear combination of vectors in } B_W$$

$$\text{Thus coordinates of } w = \mathcal{R}(v) = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix}, \text{ coordinates of } \mathcal{R}(e_j) \text{ is } c_j = \begin{bmatrix} l_{1j} \\ l_{2j} \\ \vdots \\ l_{mj} \end{bmatrix}$$

$$\text{Finally, } w_j = \sum_{j=1}^n l_{ij} v_j$$

$$w = Lv, \text{ where } L = [l_{ij}] \in \mathbb{R}^{m \times n}$$

8.4 Change of Basis

Standard Basis A standard basis for $\mathbb{R}^n$ is $\{e_1, e_2, \dots, e_n\}$ for $e_j = \begin{bmatrix} e_{1j} \\ e_{2j} \\ \vdots \\ e_{nj} \end{bmatrix}, \begin{cases} e_{ij} = 1 \text{ if } i=j \\ e_{ij} = 0 \text{ otherwise} \end{cases}$, ex $e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}$

for $\mathbb{R}^n$ is $\{e_1, e_2, \dots, e_n\}$ for $e_j = [e_{j1} \ e_{j2} \ \dots \ e_{jn}]$, $\begin{cases} e_{ij} = 1 \text{ if } i=j \\ e_{ij} = 0 \text{ otherwise} \end{cases}$, ex. $e_3 = [0 \ 0 \ 1 \ \dots \ 0]$

Changing Basis

with a matrix: If for vector space V , $B_A = \{a_1, a_2, a_3, \dots, a_n\}$ and $B_f = \{f_1, f_2, f_3, \dots, f_n\}$ are bases, then:

$\hookrightarrow$ Each vector $v \in V$ can be expressed as: $v = \sum_{i=1}^n v_i^{(a)} a_i$ or $v = \sum_{i=1}^n v_i^{(f)} f_i$

$\hookrightarrow$ Each basis vector a_k can be expressed as a linear combination of basis vectors in f :

$$a_1 = p_{11} f_1 + p_{21} f_2 + \dots + p_{n1} f_n$$

$$a_2 = p_{12} f_1 + p_{22} f_2 + \dots + p_{n2} f_n$$

$\vdots$

$$a_n = p_{1n} f_1 + p_{2n} f_2 + \dots + p_{nn} f_n$$

$$a_k = \sum_{i=1}^n p_{ik} f_i$$

linear combination of coordinates multiplied by a factor

$$\hookrightarrow \text{And, } v = \sum_{j=1}^n v_j^{(a)} a_j = \sum_{j=1}^n v_j^{(a)} \left(\sum_{i=1}^n p_{ij} f_i \right) = \sum_{i=1}^n \left(\sum_{j=1}^n p_{ij} v_j^{(a)} \right) f_i$$

$\hookrightarrow$ Finally, $v_f = P v_a$, $P = [p_{ij}] \in \mathbb{R}^{n \times n}$ is the transformation matrix

equally, $v_a = Q v_f$

Transformation Matrix: Let $\alpha = \{v_1, v_2 \dots v_n\}$ be 2 bases for V , $\dim V = n$
 $\beta = \{w_1, w_2 \dots w_n\}$

$P_{\beta\alpha} = [v_1]_{\beta} \ [v_2]_{\beta} \ \dots \ [v_n]_{\beta} \in \mathbb{R}^n$ is a **change of basis matrix from α to β**
 such that $[x]_{\beta} = P_{\beta\alpha}[x]_{\alpha}$ for every $x \in V$, $[x]_{\alpha}$ is the **coordinates vector of x in α**

Theorems of Transformation Matrices: 1.) Let $\alpha = \{v_1, v_2 \dots v_n\}$ be two bases for V , $\dim V = n$
 $\beta = \{w_1, w_2, \dots w_n\}$

then $P_{\beta\alpha}$ is invertible, $P_{\beta\alpha}^{-1} = P_{\alpha\beta} \longrightarrow P_{\alpha\beta} P_{\beta\alpha} = \bar{I}$

2.) Let α, β be 2 different bases for a finite dimensional vector space V ,
 and $T: V \rightarrow V$ be a linear transformation.

Then $[T]_{\alpha\alpha} = P_{\alpha\beta} [T]_{\beta\beta} P_{\beta\alpha} = S [T]_{\beta\beta} S^{-1}$ for $S = P_{\alpha\beta}$

Similar matrices: For $A, B \in \mathbb{R}^n$, A and B are similar if there is an invertible $S \in \mathbb{R}^n$ s.t. $A = SBS^{-1}$

$\hookrightarrow$ i.e. A, B represent the same linear transformation $V \rightarrow V$ w.r.t possibly different bases.
 If equal bases, $S = I_n$, $A = B$

Rank of similar matrices: If $A, B \in \mathbb{R}^n$ are similar, then $\text{rank } A = \text{rank } B$

9.1 Determinants of 2×2 Matrices

Definition of determinant: For a 2×2 matrix A , the function **det**: ${}^2\mathbb{R}^2 \rightarrow \mathbb{R}$ defined on rows of A :
 satisfies:

Multilinearity property: i) $\det \begin{bmatrix} ba_1 + ca'_1 \\ a_2 \end{bmatrix} = b \det \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + c \det \begin{bmatrix} a'_1 \\ a_2 \end{bmatrix}$ for $b, c \in \mathbb{R}$

Alternating property: ii) $\det \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = - \det \begin{bmatrix} a_2 \\ a_1 \end{bmatrix}$

"Determining value"
 $\hookrightarrow$ Similar to line
 defined for line = 1
 iii) $\det \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \det(I_2) = 1$

for $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ ^{rows}

$\star(i^*)$ $\det \begin{bmatrix} a_1 \\ ba_2 + ca'_2 \end{bmatrix} = b \det \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + c \det \begin{bmatrix} a_1 \\ a'_2 \end{bmatrix}$ can be shown using i, ii

Property of 2×2 determinant: 1.) If A has equal rows, $\det A = 0$

2.) If A has a 0 row, $\det A = 0$

3.) If A has linearly dependent rows, $\det A = 0$

Inverse of 2×2 matrix: For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $A^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Formula for 2x2 matrices: $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - cb$

Determinant of Transpose: If $A \in {}^2\mathbb{R}^2$ then $\det A^T = \det A$

Now in terms of columns, if $A = [c_1 \ c_2]$,

- $\det [bc_1 + c_1' \ c_2] = b \det [c_1 \ c_2] + c \det [c_1' \ c_2]$
- $\det [c_1 \ c_2] = -\det [c_2 \ c_1]$
- If A has equal columns then $\det A = 0$
- If A has a 0 column, $\det A = 0$
- If columns of A are linearly dependent then $\det A = 0$.

Matrix Minor: An ij minor of an $n \times n$ matrix is a $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j of A , denoted A_{ij}

Determinant of $n \times n$ matrices:

- $\det A = \det (A \text{ where row}_i \text{ is replaced with row}_i + \text{row}_j)$ "Row operation does not change"
- $\det (A \text{ row } r_i \text{ multiplied by } \lambda) = \lambda \det A$

Properties of determinants:

- If A has a zero row, $\det(A) = 0$
- If any row is replaced by a linear combination of all other rows, $\det A$ remains unchanged
 $\hookrightarrow \det \begin{bmatrix} r_1 \\ \vdots \\ r_i \\ \vdots \\ r_i \\ \vdots \\ r_n \end{bmatrix} = \det \begin{bmatrix} r_1 \\ \vdots \\ \sum_{k=1}^i c_k r_k \\ \vdots \\ r_n \end{bmatrix}$
- If any rows of A is linearly dependent, $\det A = 0$
- If row i is exchanged with row j , say A'
 $\det A = -\det A'$
- If $\det(I_n) = 1$

$$A = \begin{bmatrix} r_1 \\ \vdots \\ r_{i-1} \\ \lambda p + \mu q \\ r_{i+1} \\ \vdots \\ r_n \end{bmatrix}$$

$$\det A = \lambda \det \begin{bmatrix} r_1 \\ \vdots \\ r_{i-1} \\ p \\ r_{i+1} \\ \vdots \\ r_n \end{bmatrix} + \mu \det \begin{bmatrix} r_1 \\ \vdots \\ r_{i-1} \\ q \\ r_{i+1} \\ \vdots \\ r_n \end{bmatrix}$$

Determinants of 3x3 or beyond:

For $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \in {}^3\mathbb{R}^3$, $\det A = a_{11} \det A_{11} - a_{12} \det A_{12} + a_{13} \det A_{13}$

Alternating signs $\rightarrow \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$

Note equal indices

$$= -a_{21} \det A_{21} + a_{22} \det A_{22} - a_{23} \det A_{23}$$

$$= a_{31} \det A_{31} - a_{32} \det A_{32} + a_{33} \det A_{33}$$

Cofactors: $C_{ij} = (-1)^{i+j} \det A_{ij}$

Thus for $A \in \mathbb{R}^3$, $\det A = \sum_{j=1}^3 a_{ij} C_{ij}$, for $1 \leq i \leq 3$
 $A \in \mathbb{R}^n$, $\det A = \sum_{j=1}^n a_{ij} C_{ij}$

Determinant of Diagonal Matrices: For $D \in \mathbb{R}^n$, a diagonal matrix, if the diagonal components are $d_1 \dots d_n$.
Then $\det D = d_1(d_2)(d_3) \dots (d_n)$

$$= \prod_{k=1}^n d_k$$

Determinant of Upper-triangular Matrix: For $U \in \mathbb{R}^n$, an upper triangular matrix, then
 $\det U = (u_{11})(u_{22})(u_{33}) \dots (u_{nn})$

$$= \prod_{k=1}^n u_{kk}$$

9.3 Properties of determinants

Product Rule: For $A, B \in \mathbb{R}^n$
 $\det AB = \det A \det B$

Theorem of Transpose: For $A \in \mathbb{R}^n$
 $\det A^T = \det A$

Invertibility Rule: $\hookrightarrow$ If $A \in \mathbb{R}^n$.
 A is invertible iff $\det A \neq 0$

$$\hookrightarrow \text{If } A \text{ is invertible,} \\ \det A^{-1} = \frac{1}{\det A}$$

Deletion Matrix: For $R_k \in \mathbb{R}^{n-1}$, a deletion matrix, is made by removing k^{th} column of I
For example for $\mathbb{R}^3$,

$$R_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R_2^T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Usage: For $A \in \mathbb{R}^n$, $R_k \in \mathbb{R}^{n-1}$,

$\hookrightarrow AR_k$ is the matrix obtained by removing k^{th} column of A

$\hookrightarrow R_k^T A$ is the matrix obtained by removing k^{th} row of A

$$\text{For ex. } A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \\ R_2^T A R_3 = \begin{bmatrix} 1 & 2 \\ 7 & 8 \end{bmatrix}$$

9.5 Adjoints

Adjoints: For $A \in \mathbb{R}^n$,
 $\text{adj } A = C^T$, where $C = [c_{ij}] \in \mathbb{R}^n$ and c_{ij} is the (i,j) cofactor of A

Theorem of Adjoints: For $A \in \mathbb{R}^n$,

$$\hookrightarrow A(\text{adj } A) = (\text{adj } A)A = (\det A)I$$

$$\hookrightarrow \text{If } A \text{ is invertible,}$$
$$A^{-1} = \frac{1}{\det A} \text{adj } A$$

10.2 Eigen problem

Eigen problem: $Ap = \lambda p$
and $(\lambda I - A)p = 0$ or $(A - \lambda I)p = 0$ for $p \neq 0$

Where $p \neq 0$ is the eigenvector

The corresponding λ is the eigenvalue of A

Eigenspace: For a given eigenvalue λ , the eigenspace of A corresponding to eigenvalue λ is:
 $E_\lambda(A) = \text{null}(\lambda I - A)$

for example, if $\lambda = 2$ gives eigenvector $p = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ if $\lambda = 1$ gives eigenvector $p = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$
 $\hookrightarrow E_2(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$
 $\hookrightarrow E_1(A) = \text{span} \left\{ \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$

Characteristic Polynomial: For matrix A , $c_A(\lambda) \in P_n(\mathbb{R})$ is

$$c_A(\lambda) = \det(\lambda I - A)$$

To find eigenvalues: Let $\det(\lambda I - A) = 0$

Property of eigenspace: Let λ, μ be distinct eigenvalues of A ,
Then $E_\lambda \cap E_\mu = \{0\}$

10.3 Diagonalization

Diagonal Matrix: The matrix $P \in \mathbb{R}^n$ is said to diagonalize $A \in \mathbb{R}^n$ if
 P is invertible s.t.

$$P^{-1}AP = \Delta, \quad \Delta = \text{diag}\{\lambda_\alpha\}, \text{ the diagonal matrix of eigenvalues of } A$$

Properties of similar matrices: Recall A, B are similar iff $A = S B S^{-1}$ for some invertible S .

Then:

1) $\text{rank } A = \text{rank } B$

2) $\det A = \det B$

3) They have the same characteristic polynomial

4) They have the same eigenvalues

5) They have the same trace

Diagonalizability: The matrix $P \in \mathbb{R}^n$ is said to be **diagonalizable** if it is similar to a diagonal matrix $D \in \mathbb{R}^n$

Theorem of Diagonalization: Matrix $P \in \mathbb{R}^n$ diagonalizes $A \in \mathbb{R}^n$ where

$$P^{-1} A P = \Delta = \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix}, \text{ iff the columns of } P \text{ form a basis for } \mathbb{R}^n \text{ consisting of eigenvectors } p_\alpha \text{ of } A, \text{ where } A p_\alpha = \lambda_\alpha p_\alpha$$

Corollary: $A \in \mathbb{R}^n$ is diagonalizable iff $\mathbb{R}^n$ has a basis consisting of eigenvectors of A

Characteristic polynomial, determinants and traces: Let $P \in \mathbb{R}^n$ diagonalize $A \in \mathbb{R}^n$, let $\lambda_1, \dots, \lambda_n$ be eigenvalues of A ,
Then:

a) $c_A(\lambda) = c_\Delta(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_n) = \prod_{\alpha=1}^n (\lambda - \lambda_\alpha)$

b) $\det A = \det \Delta = \lambda_1 \lambda_2 \cdots \lambda_n = \prod_{\alpha=1}^n \lambda_\alpha \longrightarrow \det A = \text{product of eigenvalues of } A$

c) $\text{tr } A = \text{tr } \Delta = \lambda_1 + \lambda_2 + \cdots + \lambda_n = \sum_{\alpha=1}^n \lambda_\alpha \longrightarrow \text{tr } A = \text{sum of eigenvalues of } A$

10.4 Conditions for Diagonalizability

Distinct Eigenvalues and linear independency: Let $A \in \mathbb{R}^n$ with distinct eigenvalues $\lambda_1, \dots, \lambda_r, r \leq n$.

If $x_\alpha \in E_{\lambda_\alpha}$, for $x_\alpha \neq 0$ (eigenvector x_α belongs to eigenspace E_{λ_α})
Then $\{x_1, x_2, \dots, x_r\}$ is linearly independent.

Corollary: If $A \in \mathbb{R}^n$ has n distinct eigenvalues, then A is diagonalizable.

Lemma: Let $A \in \mathbb{R}^n$ with distinct eigenvalues $\lambda_1, \dots, \lambda_r, r \leq n$. If $x_\alpha \in E_{\lambda_\alpha}$ and

$$x_1 + x_2 + \cdots + x_r = 0$$

Then $x_\alpha = 0, \alpha = 1, \dots, r$.

Distinct eigenvalues and eigenspace: Let $A \in \mathbb{R}^n$ with distinct eigenvalues $\lambda_1, \dots, \lambda_r, r \leq n$.

If H_{λ_α} is a basis for E_{λ_α} , then:

$H = H_{\lambda_1} \cup H_{\lambda_2} \cup \dots \cup H_{\lambda_r} \equiv \bigcup_{\alpha=1}^r H_{\lambda_\alpha}$ is linear independent and
 $m_1 + m_2 + \dots + m_r \leq n$, where $m_\alpha = \dim E_{\lambda_\alpha}$

Corollary: If $A \in \mathbb{R}^n$ has distinct eigenvalues $\lambda_\alpha, \alpha = 1 \dots r$, and $m_\alpha = \dim E_{\lambda_\alpha}$ for $\alpha = 1 \dots r$ s.t.

$m_1 + m_2 + \dots + m_r = n$, then A is diagonalizable

10.5 Multiplicities of Eigenvalues

Let $A \in \mathbb{R}^n$ with eigenvalues λ_α .

Algebraic Multiplicity: The highest power of n_α of $1 - \lambda_\alpha$ that divides the characteristic polynomial s.t.
 $c_A(\lambda) = (1 - \lambda_\alpha)^{n_\alpha} g(\lambda)$ (Readable language: How many times did λ_α repeat itself in $c_A(\lambda)$)
 $\hookrightarrow$ For $n_\alpha > 1$, the eigenvalue repeats

Geometric Multiplicity: The dimension m_α of E_{λ_α} is the geometric multiplicity of λ_α .

Multiplicity Theorem: Let λ_α be an eigenvalue of $A \in \mathbb{R}^n$

Then $1 \leq m_\alpha \leq n_\alpha$, where $m_\alpha \rightarrow$ geometric, $n_\alpha \rightarrow$ algebraic multiplicities of λ_α .

$\hookrightarrow$ Particularly, if $n_\alpha = 1$, then $m_\alpha = 1$

Random proposition (?): Let $\bar{A} = \begin{bmatrix} \bar{I} & \bar{B} \\ \bar{O} & \bar{C} \end{bmatrix} \in \mathbb{R}^n$, $B \in \mathbb{R}^{n \times r}$, $C \in \mathbb{R}^{r \times r}$, $\bar{I} \in \mathbb{R}^r = \bar{I}_r$

Then $\det A = \det C$

Diagonalization test: Let $A \in \mathbb{R}^n$ with distinct eigenvalues $\lambda_\alpha, \alpha = 1 \dots r$. Then A is diagonalizable iff

$m_\alpha = n_\alpha$ for all $\alpha = 1 \dots r$, i.e. algebraic multiplicity = geometric multiplicity for all $\lambda_\alpha, \alpha = 1 \dots r$

Contrapositive: If A has any λ_α where $m_\alpha \neq n_\alpha$, A is not diagonalizable.

10.6 Linear first-order ODEs

2x2 systems: A system of the form: $x_1' = a_{11}x_1 + a_{12}x_2$ where $a_{11}, a_{12}, a_{21}, a_{22} \in \mathbb{R}$ is called a
 $x_2' = a_{21}x_1 + a_{22}x_2$

$\hookrightarrow$ 2x2 system of homogeneous linear first order differential equations with constant coefficients

$\hookrightarrow$ We can express the system as $x' = Ax$ $\longrightarrow A = [a_{ij}] \in \mathbb{R}^2$

$n \times n$ systems: A system of the form: $x_1' = a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n$
 $x_2' = a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n$
 $\vdots$
 $x_n' = a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n$, where $a_{ij} \in \mathbb{R}$ for $j \in \{1, 2, \dots, n\}$ is called a:

↳ $n \times n$ system of homogeneous linear first order differential equations with constant coefficients

↳ We can express the system as $x' = Ax$ $\longrightarrow A = [a_{ij}] \in \mathbb{R}^n$

Derivatives: In case lil' bro was unaware,

$$x' = \dot{x} = \frac{dx}{dt}$$

Existence of solution: If $A \in \mathbb{R}^n$ is diagonalizable and $x_0 \in \mathbb{R}^n$, then there is a solution of $x' = Ax$ that satisfies the initial condition $x(0) = x_0$.

General solution: Let $A \in \mathbb{R}^n$ be diagonalizable and the invertible matrix $P \in \mathbb{R}^n$ and diagonal matrix $\Delta = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) \in \mathbb{R}^n$ s.t. $A = P\Delta P^{-1}$, then the general solution to $x' = Ax$ is:

$$x(t) = c_1 e^{\lambda_1 t} p_1 + c_2 e^{\lambda_2 t} p_2 + \dots + c_n e^{\lambda_n t} p_n = \sum_{i=1}^n c_i e^{\lambda_i t} p_i$$

corresponding eigenvalue and eigenvectors

where p_i is the i th column of P , λ_i is the i th element of the diagonal of Δ , $c_i \in \mathbb{R}$ for $i \in \{1, 2, \dots, n\}$

Note: It can still be solved for $x' = Ax$ when A is not diagonalizable.

Bonus

Let A be $l \times m$, B be $m \times n$

Row picture of Matrix Multiplication:

$$AB = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_{l-1} \\ r_l \end{bmatrix} \begin{bmatrix} | & & | \\ B & & B \\ | & & | \end{bmatrix} = \begin{bmatrix} r_1 B \\ r_2 B \\ \vdots \\ r_{l-1} B \\ r_l B \end{bmatrix}$$

$l \times m$ $1 \times m$ $m \times n$ $l \times n$

Column picture of Matrix Multiplication:

$$AB = \begin{bmatrix} | & & | \\ A & & A \\ | & & | \end{bmatrix} \begin{bmatrix} c_1 & c_2 & \dots & c_{n-1} & c_n \end{bmatrix} = \begin{bmatrix} A c_1 & A c_2 & \dots & A c_{n-1} & A c_n \end{bmatrix}$$

$l \times m$ $m \times n$ $l \times n$ $m \times 1$ $l \times 1$